

Holt-Winters Forecasting Method

The Holt-Winters method, also known as triple exponential smoothing, is a forecasting technique used for time series data that contains level, trend, and seasonality.

Key Components

1. Level (L_t): The baseline value of the series at time t .
2. Trend (T_t): The growth or decline in the series over time.
3. Seasonality (S_t): The repeating seasonal pattern.

Example Data (Monthly Sales)

Month	Sales
Jan	30
Feb	21
Mar	29
Apr	31
May	40
Jun	48

Update Equations (Additive Holt-Winters)

Level: $L_t = \alpha(Y_t - S_{t-m}) + (1-\alpha)(L_{t-1} + T_{t-1})$

Trend: $T_t = \beta(L_t - L_{t-1}) + (1-\beta)T_{t-1}$

Seasonality: $S_t = \gamma(Y_t - L_t) + (1-\gamma)S_{t-m}$

Forecast Formula

Forecast h steps ahead:

$$Y_{t+h} = L_t + hT_t + S_{t-m+h}$$

Simple Numerical Illustration

Suppose the estimated components are:

$$\text{Level } (L_t) = 40$$

$$\text{Trend } (T_t) = 3$$

$$\text{Seasonal factor } (S) = 2$$

Forecast for the next period:

$$Y(t+1) = 40 + (1 \times 3) + 2 = 45$$

So the forecast value is 45.